

Exercise (Spring Context - Spring Container)

Q1

```
1 usage
@SpringBootApplication
public class SpringPollApplication {

    public static void main(String[] args) { SpringApplication.run(SpringPollApplication.class, args); }

    @Bean
    public String getMessage1(){
        System.out.println("hey from message1");
        return "1";
    }
}
```

The out put:

Hey from message1

1- The container will contain "1" as a String only

Exercise (Spring Context - Spring Container)

Q2

```
1 usage
2 @SpringBootApplication
3 public class SpringPollApplication {
4
5     public static void main(String[] args) { SpringApplication.run(SpringPollApplication.class, args); }
6
7     @Bean
8     @Qualifier("1")
9     public String getMessage1(){
10         System.out.println("hey from message1");
11         return "1";
12     }
13
14     @Bean
15     public String getMessage2(@Qualifier("1") String data ){
16         System.out.println("hey from message2");
17         return data ;
18     }
19 }
```

The out put:

hey from message1

hey from message2

- 1- Method getMessage1 will add "1" to the container
- 2- Then getMessage2 will take "1" as parameter according to @Qualifier("1")
- 3- Method getMessage2 will ad (data) to the container

Exercise (Spring Context - Spring Container)

Q3

```
@Bean
@Qualifier("1")
public String getMessage1(){
    System.out.println("hey from message1");
    return "1";
}

@Bean
@Qualifier("2")
public String getMessage2(@Qualifier("3") String data ){
    System.out.println("hey from message2");
    return data;
}

@Bean
@Qualifier("3")
public String getMessage3(){
    System.out.println("hey from message3");
    return "3" ;
}
```

Out put:

hey from message1

hey from message3

hey from message2

- 1- getMessage1 will be printed
- 2- getMessage1 @Qualifier will not do anything bc it's not used
- 3- getMessage3 will be executed after bc getMessage2 depends on it
- 4- getMessage3 will add "3" to the container
- 5- getMessage2 will take "3" as parameter
- 6- getMessage2 will add (data) to the container

Out put:

hey from message3

hey from message1

hey from message2

- 1- getMessage3 executes and adds "3" to the container.
- 2- getMessage1 executes; its @Qualifier is not used.
- 3- getMessage2 takes "3" as data and adds it to the container.

Out put:

hey from message3

hey from message2

hey from message1

- 1- getMessage3 executes and adds "3" to the container.
- 2- getMessage2 takes "3" as data and adds it to the container.
- 3- getMessage1 executes; its @Qualifier is not used.

Exercise (Spring Context - Spring Container)

Q4

```
@Bean
@Qualifier("1")
public String getMessage1(){
    System.out.println("hey from message1");
    return "1";
}

@Bean
@Qualifier("2")
public String getMessage2(@Qualifier("3") String data ){
    System.out.println("hey from message2");
    return data;
}

@Bean
@Qualifier("3")
public String getMessage3(){
    System.out.println("hey from message3");
    return "3" ;
}

@Component
public class MainController {

    1 usage
    String data;

    public MainController(@Qualifier("1") String data){
        this.data=data;
        System.out.println("hey from Main controller");
    }
}
```

Out put:

hey from message1

hey from Main controller

hey from message3

hey from message2

- 1- getMessage1 will be printed, and send "1" to the container
- 2- getMessage1 @Qualifier will be received as parameter in the MainController constructor
- 3- Controller Message will be printed
- 4- getMessage3 will be executed next, it's message will printed, it will send "3" in the container
- 5- "3" will be received by getMessage2 as parameter
- 6- getMessage2 will print it's message and it will send (data) to the container

Out put:

hey from message3

hey from message1

hey from message2

hey from Main controller

- 1- getMessage3 executes and sends "3" to the container.
- 2- getMessage1 executes and sends "1" to the container.
- 3- getMessage2 takes "3" as a parameter and executes.
- 4- MainController takes "1" as a parameter and executes.

Out put:

hey from message1

hey from message3

hey from message2

hey from Main controller

- 1- getMessage1 executes and sends "1" to the container.
- 2- getMessage3 executes and sends "3" to the container.
- 3- getMessage2 takes "3" as a parameter and executes.
- 4- MainController takes "1" as a parameter and executes.

Exercise (Spring Context - Spring Container)

Q5

```
15
16 @Bean
17 @Qualifier("1")
18 public String getMessage1(MainController mainController){
19     System.out.println("hey from message1");
20     return "1";
21 }
22
23 @Bean
24 @Qualifier("2")
25 public String getMessage2(@Qualifier("3") String data ){
26     System.out.println("hey from message2");
27     return data;
28 }
29
30 @Bean
31 @Qualifier("3")
32 public String getMessage3(){
33     System.out.println("hey from message3");
34     return "3" ;
35 }
36
37
38 import org.springframework.beans.factory.annotation.Qualifier;
39 import org.springframework.stereotype.Component;
40
41 @Component
42 public class MainController {
43
44     String data;
45
46     public MainController(@Qualifier("2") String data){
47         this.data=data;
48         System.out.println("hey from Main controller");
49     }
50 }
```

Out put:

hey from message3

hey from message2

hey from Main controller

hey from message1

- 1- First it's start with getMessage3 bc it didn't depends on any parameter, it will print it's message then add "3" to the container
- 2- According to the @Qualifier("3") will go to getMessage2 and it will be executed, it's message will printed, (data) will be added to the container
- 3- bc getMessage1 depends on the Controller the compiler will go to Controller class
- 4- in Controller Class the constructor depends on getMessage2 according on @ Qualifier("2"), and it will receive data, and send it to getMessage1 method
- 5- getMessage1 will take (data) from the controller as parameter, and prints it's message, "1" will be added to the container